Hemal Arora

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EDUCATION

Stanford University **Stanford, CA**

M.S. Electrical Engineering (Signal Processing, Control and Optimization) *Expected Dec 2027*

B.S. Electrical Engineering (Hardware & Software Track) | **GPA:** 3.85/4.0 *Expected Jun 2027*

- **Coursework:** Robot Dynamics, Manipulation & Control (CS223A/CS225A/AA174A); Robot Perception & Computer Vision (CS227A/CS231N); Reinforcement Learning (CS234/CS238); Machine Learning & Statistical Inference (CS229/EE278); Computer Systems, Digital Design & VLSI (CS107E/EE180/EE271)

EXPERIENCE

Stanford Interactive Perception and Robot Learning Lab (IPRL) **Stanford, CA**

Undergraduate Researcher: Advised by Prof. Jeannette Bohg *Mar 2026 – Present*

- Co-developed ClothAtlas (submitted to ICRA 2027), a graph-conditioned flow-matching model for 3D deformable state estimation that reconstructs complete garment geometry from partial point clouds, preserves correspondences under occlusion, and generalizes to unseen mesh topologies without retraining.
- Transferred ClothAtlas directly from simulation to real-world garment manipulation without fine-tuning, localizing 85.1% of occluded grasp targets within 10 cm vs. 36.2% for dense descriptor matching.
- Researching failure-aware robot learning methods that retrieve relevant prior interactions from memory and use action-conditioned world models to guide test-time recovery in Vision-Language-Action (VLA) policies.

Mosaic Intelligence Labs (Surgical Robotics) **San Francisco, CA**

Member of Technical Staff: Simulation & Perception *Jun – Sep 2026*

- Built X-ray and Intravascular Ultrasound (IVUS) simulation pipelines using CUDA and OptiX for an endovascular robotic navigation system; calibrated against clinical data and generated ~110K synthetic frames for model training.
- Researched model architectures and optimized training procedures for surgical instrument and blood-vessel segmentation, achieving 0.94 Dice on real IVUS images; demonstrated sim-to-real transfer during an animal study.
- Extended GPU-accelerated XPBD Cosserat-rod solvers with parallel batched rod solving, coaxial instrument coupling, and stability fixes; shared technical improvements with NVIDIA's Isaac for Healthcare team for future releases.
- Built a real-time surgical navigation simulator, coupling instrument dynamics with simulated imaging for hardware-in-the-loop validation of factor-graph-based state estimation algorithms under partial observations.

Multiply Labs (Robotic Biomanufacturing) **San Francisco, CA**

Robotics Software Engineer *Jun – Dec 2025*

- Trained Diffusion Policy and Action Chunking Transformer (ACT) models on ~150 teleoperated demonstrations for precise cartridge insertion into millimeter-scale slots, deployed and evaluated on a UR10e robot.
- Developed a motion planning system for the company's robotic cell therapy manufacturing cluster in Isaac Sim, using cuMotion to execute collision-free trajectories without hard-coded waypoints.
- Built a production-ready camera SDK supporting cluster-wide ArUco auto-calibration and Unitree G1 humanoid demonstration data collection for VLA fine-tuning.

GE HealthCare & Stanford Magnetic Resonance Systems Research Lab **Stanford, CA**

Undergraduate Researcher: Advised by Prof. John Pauly *Jun – Sep 2024*

- Designed a 24 GHz mm-wave radar system for non-contact cardiac/respiratory gating in MRI, including custom PCB fabrication and SSA/ICA signal processing; first-author abstract accepted and presented at ISMRM 2025 [Abstract].

SELECTED PROJECTS & AWARDS

- **Autonomous Pickleball-Playing Mobile Manipulator [Video]:** Built a Franka FR3 + TidyBot++ mobile manipulator that tracks incoming balls, predicts intercepts, repositions its base, and replans velocity-controlled strikes at 100 Hz using damped least-squares IK and velocity-constrained Hermite trajectories.
- **Maximum Likelihood RL for Robotic Manipulation [Report]:** Extended MaxRL to sparse-reward 7-DOF manipulation, benchmarking against RLOO using goal-conditioned behavior cloning, hindsight experience replay, and policy-gradient fine-tuning.
- **1st Place, OpenAI Prize—TreeHacks 2025 [Devpost]:** Built a real-time EEG-adaptive generative storytelling system with GPT-4o, asynchronously orchestrating narrative, audio, and video generation.

SKILLS

- C++, Python, C, PyTorch, ROS 2, MuJoCo, NVIDIA Isaac Sim, Warp, Newton, OptiX, SystemVerilog, KiCad